This Exercise Report shows you how your genetic data matches up with articles on athleticism and muscle composition. Please read the full article for details and references.

Article: Fat Burning

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
PPARD	rs2267668	--	G	0.16	Less benefit from aerobic exercise; lower skeletal muscle mitochondrial function
PPARD	rs1053049	TT	C	0.25	Not as great of response to exercise for weight loss
PPARD	rs2016520	TT	C	0.2	Lower fasting plasma glucose; decreased cardiovascular disease risk

Article: Muscle Pain

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
AMPD1	rs17602729	GG	A	0.11	AMP deaminase deficiency - more likely to be sore after workout

Article: Muscle Type

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
ACTN3	rs1815739	--	T	0.43	TT = non-functioning ACTN3; more likely to be an endurance athlete than power athlete

Article: Exercise Intensity

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
ANKK1	rs1800497	GA	A	0.2	Lower exercise reward; motivation
CNR1	rs6454672	--	T	0.14	TT only: greater exercise tolerance
LEPR	rs12405556	GG	T	0.25	Greater tolerance for intense exercise
GABRA3	rs8036270	--	G	0.42	Greater tolerance for intense exercise
BDNF	rs6265	CC	T	0.18	more intrinsically motivated to exercise

Article: Athletic performance

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
AGTR2	rs11091046	AA	C	0.52	More slow-twitch muscle; endurance athlete
AGT	rs699	GA	G	0.45	Increased angiotensin; more geared towards power athlete
IL6	rs1800795	GG	C	0.34	Decreased IL6; more often found in endurance athletes
MSTN	rs1805086	TT	C	0.02	Greater muscle mass
NOS3	rs2070744	TT	C	0.34	Decreased eNOS
AMPD1	rs17602729	GG	A	0.11	AMPD1 deficiency; more muscle soreness
TTN	rs10497520	TT	T	0.17	more common in sprinters and power athletes
ACTN3	rs1815739	--	T	0.43	TT = non-functioning ACTN3; more likely to be an endurance athlete than power athlete
ACE	rs4343	AG	G	0.49	G/G = ACE deletion, no performance advantage for athletes

Article: Tendonitis

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
COL1A1	rs1800012	CC	A	0.16	A/A: significantly reduced risk of Achilles' tendon tear (good)
COL5A1	rs12722	TC	T	0.49	higher risk of Achilles' tendinopathy (lower levels of COL5A1); increased risk of tennis elbow
MIR608	rs4919510	--	G	0.21	Associated with chronic Achilles' tendinopathy
BMP4	rs2761884	--	T	0.4	Increased risk of tendinopathies
FCRL3	rs7528684	AA	G	0.45	Increased risk of tendinopathies
TNF	rs1800629	GG	A	0.15	Higher TNF-alpha levels; increased risk of Achilles tendon problems and knee tendon problems
MMP13	rs2252070	--	T	0.68	Increased risk of posterior tibial tendon problems (flat foot)
MMP1	rs1144393	--	C	0.35	Increased risk of posterior tibial tendon problems (flat foot)
MMP3	rs650108	--	G	0.72	G/G: increased risk of tendinopathies (athletes)
MMP3	rs679620	CC	T	0.48	T/T: increased risk of tendinopathies (athletes)
GDF5	i6011281	--	A	0.57	A/A: higher risk of achilles tendon (common genotype)
GDF5	rs143383	GA	A	0.57	A/A: higher risk of achilles tendon (common genotype)

[Article: Weight Loss from Aerobic Exercise]

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
BDNF	rs4923460	GG	G	0.79	Associated with decreased BMI with aerobic exercise (combined with other variants)
BDNF	rs6265	CC	T	0.18	T/T and C/T: No BMI benefit with aerobic exercise; C/C: Associated with decreased BMI with aerobic exercise (combined with other variants)
CHRNA5	rs16969968	GG	A	0.32	Associated with decreased BMI with aerobic exercise (combined with other variants)
COL1A1	rs1800012	CC	A	0.16	Associated with decreased BMI with aerobic exercise (combined with other variants)
CPT1B	rs5770917	TT	C	0.06	Associated with no decrease in BMI with aerobic exercise (combined with other variants)
FGF5	rs16998073	AA	T	0.26	Associated with decreased BMI with aerobic exercise (combined with other variants)
POU3F2	rs12206087	--	A	0.42	A/A: Not associated with decreased BMI with exercise; A/G or G/G: Associated with decreased BMI with aerobic exercise (combined with other variants)
PPARGC1A	rs3774923	--	C	0.95	Associated with decreased BMI with aerobic exercise (combined with other variants)
PPARGC1A	rs10028665	--	T	0.24	Associated with decreased BMI with aerobic exercise (combined with other variants)
SMAD3	rs17228058	AA	G	0.11	Associated with decreased BMI with aerobic exercise (combined with other variants)
TBX3	rs2384550	GA	A	0.34	Associated with decreased BMI with aerobic exercise (combined with other variants)
TDRD9	rs10149470	--	G	0.52	Associated with decreased BMI with aerobic exercise (combined with other variants)

Article: Resistance Training

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
ACTN3	rs1815739	--	T	0.43	T/T: non-functioning ACTN3 gene, more likely to be an endurance athlete than power athlete
IL15RA	rs2296135	--	C	0.53	greater muscle gain with weight training
IL15RA	rs2228059	GG	T	0.49	greater baseline muscle volume in men but lower muscle quality before training
MSTN	rs1805086	TT	C	0.02	associated with greater muscle mass; significantly greater muscle mass in African Americans; greater muscle mass in older women
PPARGC1A	rs8192678	CC	T	0.32	lower PPARGC1A activity; impairment of aerobic capacity; some advantage for strength training
CNTF	rs1800169	GG	A	0.14	non-functional CNTF; lower grip strength; lower muscle mass even with training
TRHR	rs7832552	CT	T	0.31	greater lean muscle mass; increased muscle protein formation; less muscle loss in aging
TRHR	rs16892496	AC	C	0.26	greater lean muscle mass
MTOR	rs2295080	--	G	0.4	slightly lower mTOR mRNA levels; lower frequency in power (strength training) athletes
GLI3	rs10263647	--	C	0.51	increased muscle mass (quadriceps) with resistance training
IGF2	rs680	CC	C	0.74	increased muscle, more likely to be power athlete, better sprint performance, more back muscle strength
NR3C1	rs10482616	--	C	0.84	more strength gained with training
NR3C1	rs4634384	--	T	0.49	greater muscle gains with training
NR3C1	rs6189	CC	T	0.02	glucocorticoid (cortisol) resistance; possibly lower cardiovascular risk; more lean muscle mass
NR3C1	rs6190	CC	T	0.022	glucocorticoid (cortisol) resistance; more lean muscle mass
ACE	rs4343	AG	G	0.49	ACE deletion/deletion – greater gains in resistance training
HIF1A	rs11549465	CC	T	0.1	increased HIF-1a; greater gains following endurance training; more likely to be an elite athlete
AGT	rs699	GA	G	0.45	increased angiotensin, increased risk of high blood pressure, more likely to be power athlete than endurance athlete
BDNF	rs6265	CC	T	0.18	decreased BDNF; referred to in studies as Met/Met; less BDNF released after weight training

[Article: Lactate]

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
LDHA	rs2403254	TC	T	0.48	decreased alpha-hydroxyisovalerate levels, slightly decreased LDHA function
LDHA	rs2896526	AA	G	0.16	higher baseline serum amyloid A, possibly due to reduced LDHA
SLC16A1	rs1049434	TA	A	0.41	more lactate transport; lower blood lactate levels when exercising
SLC16A1	rs7169	--	G	0.41	likely more lactate transport in muscles, less likely to have muscle injuries
SLC16A7	rs3763980	AA	T	0.24	part of a haplotype associated with elevated serum lactate in elite sprinters
SLC16A7	rs10506399	--	A	0.24	lower MCT2 expression, decrease sperm motility and sperm count
SLC16A8	rs8135665	CC	T	0.2	increased relative risk of age-related macular degeneration (AMD)
GCKR	rs1260326	CC	T	0.4	higher lactate levels, lower fasting glucose levels, less gluconeogenesis
AMPD1	rs17602729	GG	A	0.11	higher lactate levels after exercise; loss of function variation for AMP Deaminase, increased muscle soreness after exercise
PPP1R3B	rs9987289	AG	A	0.09	higher lactate levels, lower fasting glucose levels

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